



Interfaces with Other Disciplines

Does the default pecking order impact systemic risk? Evidence from Brazilian data



Michel Alexandre^{a,*}, Thiago Christiano Silva^b, Krzysztof Michalak^c,
Francisco Aparecido Rodrigues^d

^a Research Department, Central Bank of Brazil and Department of Economics, University of São Paulo, Av. Paulista 1804, São Paulo 01310-922, Brazil

^b Research Department, Central Bank of Brazil and Universidade Católica de Brasília, SBS Quadra 3 Bloco B, Brasília 70074-900, Brazil

^c Department of Information Technologies, Institute of Business Informatics, Wrocław University of Economics, Komandorska 118/120, Wrocław 53-345, Poland

^d Institute of Mathematics and Computer Science, University of São Paulo, Av. Trab. São Carlsense 400, São Carlos 13566-590, Brazil

ARTICLE INFO

Article history:

Received 30 June 2022

Accepted 23 January 2023

Available online 25 January 2023

Keywords:

Finance

Systemic risk

Default pecking order

Complex networks

ABSTRACT

The *DebtRank* approach is a popular metric used to estimate systemic risk in network models. It assumes that the loss distribution of a distressed debtor among its creditors follows a pro-rata fashion, proportional to the loan granted to the debtor. However, the banking literature documents that the loss distribution can be heterogeneous across different creditors. In this study, we extend the *DebtRank* framework assuming a heterogeneous pattern of the loss distribution, inspired by the *default pecking order*. Grounded on the empirical literature, we assume distressed debtors employ some criterion (equity, equity-to-assets ratio, out-degree, or loan extended) to rank the creditors they are willing to default on first. Applying this framework to an extensive Brazilian data set, we find that the systemic risk is sensitive to the heuristic adopted for the default pecking order, with the pro-rata pattern serving as a lower bound. We can interpret this result in light of the dual role of financial networks, which can serve as a conduit for both risk-sharing and shock propagation. DPO may affect the balance between these two effects. We find empirical evidence corroborating this hypothesis by assessing the role of interconnectedness (as measured by the network density) in driving systemic risk. When losses are mostly transmitted to the more fragile creditors, the density positively affects systemic risk. In this instance, the financial network serves primarily as a channel for shock propagation rather than risk-sharing.

© 2023 Elsevier B.V. All rights reserved.

1. Introduction

This paper aims to shed light on the following question: does the default pecking order – that is, a criterion for deciding which creditors to default on first – have some effect on the systemic risk? Network models have been extensively applied to the assessment of systemic risk in financial systems (Acemoglu, Ozdaglar, & Tahbaz-Salehi, 2015; Battiston, Gatti, Gallegati, Greenwald, & Stiglitz, 2012a; Battiston, Puliga, Kaushik, Tasca, & Caldarelli, 2012b; Bonaccolto, Caporin, & Maillet, 2022; Caccioli, Catanach, & Farmer, 2012; Eisenberg & (NOE), 2001; Gai & Kapadia, 2010; Gupta, Wang, & Lu, 2021; Hałaj & Kok, 2013; Nier, Yang, Yorulmazer, & Alentorn, 2007; Roukny, Bersini, Pirotte, Caldarelli,

& Battiston, 2013; Tang, Tong, & Xun, 2022; Upper, 2011). Here, we build on network models in which agents are interconnected through contractual debt obligations (e.g., an interbank market or a bank-firm credit network). Negative shocks (which may be idiosyncratic or endogenous to the model) are at least partially absorbed by the agent's equity. However, part of this loss may be borne by their creditors, as this shock may lead the agent to not fully honor its debt obligations with its counterparties. Systemic risk is measured as the fraction of aggregate equity lost as a result of a shock spread throughout the network.

How shocks on a given agent affect its creditors varies according to the approach. In network models, there are two approaches of shock propagation. In the Eisenberg & (NOE) (2001) (E-N) approach, contagion is triggered by the default of the agent – i.e., by the complete depletion of its resources. The loss borne by the creditors of the defaulted agent is either equal to the residual between the shock and the agent's equity (Eisenberg & (NOE), 2001; Elsinger, Lehar, & Summer, 2006; Nier et al., 2007) or corre-

* Corresponding author.

E-mail addresses: michel.alexandre@bcb.gov.br (M. Alexandre), thiago.silva@bcb.gov.br (T.C. Silva), krzysztof.michalak@ue.wroc.pl (K. Michalak), francisco@icmc.usp.br (F.A. Rodrigues).

sponds to a fraction – the *loss given default* (LGD) – of their exposures to the defaulted agent (Asarnow & Edwards, 1995; Degryse, Nguyen et al., 2007; Gai & Kapadia, 2010). The *DebtRank* (Bardoscia, Battiston, Caccioli, & Caldarelli, 2015; Battiston et al., 2012a; Battiston et al., 2012b) approach holds two main differences with respect to the E-N approach: the losses which trigger contagion are i) potential rather than real, and ii) may be partial rather than necessarily complete. This methodology poses that losses suffered by a given creditor are proportional to the fraction of equity lost by the debtor – the *distress*.

Despite the differences among them, all models reviewed above share a common characteristic: a debtor affected by a negative shock distributes the loss among its creditors following a pro-rata fashion. The loss suffered by the creditor as a percentage of its exposure on the debtor hit by the shock is fixed. The LGD is assumed to be exogenous and constant across creditors, even when a range of LGD values is tested. In the residual loss approach, the LGD can be implicitly computed as the residual loss over the debtor's total liabilities. In the *DebtRank* approach, the loss transmitted by the stressed debtor as a fraction of their creditors' claims is equal to the fraction of the equity lost by that debtor. That is, it is equal to the loss suffered by the debtor as a fraction of its equity.

However, the assumption that losses are imposed on creditors proportionally to the loan extended to the debtor under distress is unrealistic. It is more plausible to assume that debtors will choose to default on certain creditors first before transmitting the residual loss to other creditors. Agents can be better off by setting a *default pecking order* (DPO) rather than adopting a pro-rata mechanism for loss distribution. There is evidence of *selective default*, when debtors discriminate among their creditors, by different economic agents, such as firms (Schiantarelli, Stacchini, & Strahan, 2020), households (Li, Liu, & Deininger, 2013; Li, Yanyan, & Deininger, 2009), governments (D'Erasmo & Mendoza, 2021), and banks.¹

The rationale behind the adoption of the DPO is that default implies the impairment of the relationship with the creditor. For instance, creditors will grant fewer loans to borrowers who have defaulted on them in the past. Therefore, agents will be more likely to default on weaker creditors due to the smaller expected value of continuing this relationship (Schiantarelli et al., 2020). In the model developed by Bertschinger, Hoefer, & Schmand (2019), agents maximize the resources that propagate through the network and return as additional internal assets. Under this criterion, the pro-rata mechanism is not always optimal. Thus, in a more realistic framework, each creditor would lose a fraction of its loans depending on a set of creditor's attributes used by debtors to set the DPO.

In this study, we extend the *differential DebtRank* approach (Bardoscia et al., 2015) by assuming a heterogeneous distribution of losses by distressed debtors. In our framework, as in the *differential DebtRank* methodology, contagion is triggered by stress – i.e., a partial loss of equity. The aggregate loss imposed by a given debtor on its counterparties equals the fraction of equity lost by the debtor times its liabilities. Although this mechanism resembles that of the *differential DebtRank* approach, there is a key difference between that framework and ours. In the *differential DebtRank* approach, losses are due to a mark-to-market adjustment made by the creditors of their exposures on the stressed debtor. On the other hand, we assume distressed debtors will perform a *strategic default*. This term was coined by bankers to define a default situation in which there is the ability, but not the willingness, to pay (Das & Meadows, 2013). This is not an uncommon phenomenon.

For instance, assessing the U.S. housing mortgage market, Guiso, Sapienza, & Zingales (2013) found out 26.4% (35.1%) of defaults appear to be strategic in March 2009 (September 2010). The few empirical studies addressing the determinants of strategic default on corporate loans (Asimakopoulou, Avramidis, Malliaropoulou, & Travlos, 2016; Karthik, Subramanyam, Shrivastava, & Joshi, 2018) bring empirical evidence that a decrease in the debtor's profitability – i.e., a worsening in its financial health – increases its probability of strategic default. This is coherent with our hypothesis that distressed debtors will perform a strategic default.

The debtor will choose a heuristic to rank their creditors, setting a DPO for the loss distribution. We will set heuristics based on two pieces of evidence brought by the literature:

1. **Creditor's weakness and default probability.** The first evidence is that creditor's weakness negatively affects its probability of default. Results presented by Schiantarelli et al. (2020) show, everything else constant, debtors are more likely to default on weaker creditors. The expected value of continuing a relationship with a weaker creditor is smaller, as it has a smaller ability in fulfilling the debtor's needs (e.g., its demand for credit). An agent's strength can be proxied by its size. Some studies on the creditor-specific determinants of loan default (Hu, Yang, & Yung-Ho, 2004; Rajan & Dhal, 2003; Salas & Saurina, 2002) mention size as negatively correlated to non-performing loans. This can be explained by the fact that large banks are more involved in risk diversification (Rajan & Dhal, 2003; Salas & Saurina, 2002) and have higher capabilities for loan evaluation (Hu et al., 2004). However, another possible explanation is that debtors are more likely to default on loans extended by weaker creditors first for the reasons discussed above.²
2. **Peer effects on default decisions.** The second evidence is that there are peer effects on debtor's default decisions. This is discussed in some theoretical frameworks using global games (Bond & Rai, 2009; Carrasco & Salgado, 2014; Drozd & Serrano-Padial, 2018). Moreover, there is evidence of peer effects on default decisions brought by empirical studies (Breza, 2012; Li et al., 2013; Li et al., 2009). Similar results were provided by the experimental study of Trautmann & Vlahu (2013). Default peer effects work through many channels. An increase in the number of defaulting borrowers may, for instance, i) threaten the creditor's future lending ability (Bond & Rai, 2009), ii) increase the lender's verification cost (Carrasco & Salgado, 2014), iii) convey information about the probability of being sued (Guiso et al., 2013), and iv) reduce the lender's enforcement ability (Drozd & Serrano-Padial, 2018; Vlahu, 2008). All these channels provide an incentive for other borrowers to default as well.

Based on these evidences, we set four heuristics. Relying on point (1), we assume the distressed debtor ranks its creditors in ascending order according to their strength. Creditor's strength is measured by two variables: total equity and equity-to-assets ratio. The debtor's loss will be transmitted to the first creditor of the list – that is, the weakest creditor –, at the limit of the loan extended by this creditor to the debtor. The residual loss, if any, will be transmitted to the next creditor in the list, and so on, until all the debtor's loss has been transmitted to its creditors. Relying on the default peer effects – point (2) –, we assume default incentives are driven by the creditor's out-degree (i.e., its number

¹ For instance, the Lehman Brothers' liquidation plan, filed on January 25, 2011, redistributed the payouts made to some important creditors (Summe, 2012).

² There are counterbalancing effects of creditors' size on their non-performing loans. The "too-big-to-fail" channel can lead large banks to engage in riskier activities, as they expect to be protected by the government in case of failure (Stern & Feldman, 2004). Hence, the overall effect of creditor's size on non-performing loans is ambiguous. At any rate, it does not invalidate our hypothesis, as we are discussing to which creditors the debtor will transmit its losses *given that it has already decided to default*.

of debtors). As discussed above, the probability of a given borrower defaulting on a given creditor is negatively related to some attributes of the creditor (e.g., its future viability as a lender or its ability to enforce defaulting loans). Moreover, borrowers default negatively affects these attributes. Suppose the borrower decides to default if these attributes fall below a given threshold. These attributes would not be threatened by a few defaults if the creditor has a high out-degree. On the other hand, a debtor would be more likely to default on a creditor with a small out-degree, as in this case, a few defaults could be enough to reduce the attributes to a level below the threshold. We also consider a heuristic in which debtors rank their creditors according to the loan granted by them. As the default leads to the impairment of the relationship with the creditor, it is reasonable to assume that debtors will be willing to default first on those creditors that granted them smaller loans. Finally, it is important to verify whether the impact on the systemic risk – if any – is due to the heterogeneous distribution of losses *per se* or to the heterogeneity in the loss distribution according to the heuristics we set. To address this point, we set a heuristic according to which creditors are randomly ranked by distressed debtors.

We apply this framework to a data set comprising quarterly information (from March 2012 through December 2015) on two Brazilian credit networks: the bank-bank (interbank) network and the firm-bank bipartite network. On the combination of these two layers, we compute the systemic risk for different levels of the initial shock. The shock is represented by an equity loss suffered by institution i in a given fraction ζ . The fraction of the overall system equity lost due to the propagation of this shock throughout the network is the systemic risk $s_{i,\zeta}$. The systemic risk is computed according to three rules of loss distribution: i) homogeneous distribution (the standard one), ii) heterogeneous distribution according to each of the four heuristics (equity, equity-to-assets ratio, out-degree, and loans extended), and iii) heterogeneous distribution following a random sorting of the creditors.

Our results show the adoption of a DPO considerably increases the systemic risk *vis-à-vis* the standard approach. For an initial shock of 0.1, the random sorting entails a systemic risk 7.4 (15.5) times greater than the homogeneous distribution of losses in the case of banks (firms). According to the heuristics, the sorting entails an even greater increase of systemic risk, varying between 11.4 and 27.5 (32.2 and 94.8) times higher in the case of the banks (firms). It shows the heuristic also plays an important role in increasing systemic risk. When loss is transmitted preferentially to more fragile agents rather than randomly, the average systemic risk is higher. Moreover, the proportional increase in the systemic risk is higher for smaller values of the initial shock. For an initial shock of 0.5, in the case of banks, the ratio between the systemic risk entailed by the heterogeneous distribution of losses and that entailed by the standard approach is between 3.0 and 7.3. If the initial shock is equal to 1 (i.e., complete default), these ratios are even smaller: between 2.4 and 4.6. A similar pattern can be observed for the firms. Finally, we also find the rise in the systemic risk brought by the heterogeneous distribution over the homogeneous case is higher when the shock is on small-sized agents.

Our findings corroborate those of Tran, Vuong, & Zeckhauser (2018). In a stylized banking network, the authors show that sequential losses entail a smaller total loss to the system than a single larger loss of the same cumulative magnitude. In our study, we show that the systemic risk is greater when the loss is concentrated in a few creditors instead of being shared equally among all creditors. Our results are also related to the dual role played by financial networks in its *robust-yet-fragile* nature (Chinazzi & Fagiolo, 2015). They are a channel for risk-sharing but also propagate shocks. When a proportional distribution of losses is adopted, as in the standard approach, there is an equilibrium between the two ef-

fects. On the other hand, if the loss is mainly transmitted to more fragile creditors, shock propagation prevails over risk-sharing, and the systemic risk is higher.

We test this hypothesis by assessing the determinants of systemic risk. Among the potential explanatory variables, we include interconnectedness, as measured by the density of the financial network. We perform this task employing machine learning techniques (random forest and XGBoost) and Shapley values analysis. The results corroborate this hypothesis. Under the homogeneous loss distribution, the impact of interconnectedness on systemic risk is meager. However, when some criteria to rank creditors are adopted, the density positively impacts the systemic risk. It suggests that, in the first case, the two financial network effects (risk-sharing and shock propagation) counterbalance. In the second case, the shock propagation effect gains relative importance over the risk-sharing effect. Under the DPO, the weakest nodes bear a fraction of the loss greater than their share of the loan extended to distressed debtors. Thus, more interconnections would result in a heavier penalty on them and lead to a higher systemic risk.

Our contribution to the literature is threefold. First, we develop a new methodology to compute the systemic risk, incorporating a more empirically plausible assumption – namely, that distressed debtors prefer to default on the weaker creditors first rather than distribute the losses proportionally to the loan granted. Second, we show the unrealistic assumption of homogeneous loss distribution leads to a non-negligible underestimation of the systemic risk. Therefore, by incorporating our methodology, systemic risk models can provide a more accurate measure of systemic risk. Finally, we shed some light on the role played by interconnectedness in driving systemic risk. We show this depends on the assumptions concerning the loss distribution by distressed debtors. Essentially, when losses are mainly transmitted to weaker creditors, a more interconnected network leads to a higher systemic risk.

This paper proceeds as follows. The data set and methodological issues are discussed in Sections 2 and 3, respectively. Section 4 brings the results. Finally, final considerations are presented in Section 5.

2. The data set

Our data set comprises several unique Brazilian databases with supervisory and accounting data. We extract quarterly information from March 2012 through December 2015 (16 periods) and build two networks: the bank-bank (interbank) network and the bank-firm bipartite network.

In the interbank network, we consider all types of unsecured financial instruments registered in the Central Bank of Brazil. The main types of financial instruments are credit, capital, foreign exchange operations, and money markets. These operations are registered and controlled by different custodian institutions: Cetip³ (private securities), the Central Bank of Brazil's Credit Risk Bureau System (SCR)⁴ (credit-based operations), and

³ Cetip is a depositary of mainly private fixed income, state and city public securities, and other securities. As a central securities depositary, Cetip processes the issue, redemption, and custody of securities and, when applicable, the payment of interest and other related events. The institutions eligible to participate in Cetip include commercial banks, multiple banks, savings banks, investment banks, development banks, brokerage companies, securities distribution companies, goods, and future contracts brokerage companies, leasing companies, institutional investors, non-financial companies (including investment funds and private pension companies) and foreign investors.

⁴ SCR is a very thorough data set that records every single credit operation within the Brazilian financial system worth 200BRL or above. Up to June 30th, 2016, this lower limit was 1,000BRL. Therefore, all the data we assess have been retrieved under this rule. SCR details, among other things, the identification of the bank, the client, the loan's time to maturity and the overdue parcel, modality of loan, credit

Table 1
Topological features of the financial networks – average over periods.

Variable	Interbank network	Bank-firm credit network
Number of banks	128.75	128.75
Number of firms	–	313.50
Banks' in/out-degree – average	10.17	22.23
Banks' in-degree – minimum	0	–
Banks' in-degree – maximum	49.81	–
Banks' out-degree – minimum	0	1
Banks' out-degree – maximum	80.38	233.94
Firms' in-degree – average	–	4.99
Firms' in-degree – minimum	–	1
Firms' in-degree – maximum	–	27.19
Assortativity	-0.3652	-0.3649
Density	0.0793	0.0581

the BM&FBOVESPA⁵ (swaps and options operations). On March 30th, 2017, BM&FBOVESPA and Cetip merged into a new company named B3.

We consider net financial exposures among different financial conglomerates or individual financial institutions that do not belong to conglomerates (classified as “b1”, “b2”, or “b4” in the Central Bank of Brazil's classification system), removing intra-conglomerate exposures. Institutions with negative equity were excluded. Financial institutions' equity was retrieved from <https://www3.bcb.gov.br/infdata>.

In the bank-firm network, we considered accounting and supervisory data from non-financial firms listed on the Brazilian stock exchange (BM&FBOVESPA). Information on firms' equity was retrieved from the *Economática* database. For each of these firms, we identified the loans granted by financial institutions using the SCR information. The criteria to include a financial institution in the bank-firm network are the same of the interbank network – that is, financial institutions with positive equity and classified as “b1”, “b2”, or “b4”.

Table 1 brings some statistics of the financial networks. They present some characteristics reported by other empirical studies on financial networks, such as disassortative behavior (Bottazzi, De Sanctis, & Vanni, 2020), sparseness (de Souza, Silva, Tabak, & Guerra, 2016), and a distribution of banks' degrees wider than that of firms in the bank-firm network (Luu & Lux, 2019). In each period t , we combine both networks to create the overall matrix of exposures $\mathbf{A}_t \in \mathbf{NB}_t \times (\mathbf{NB}_t + \mathbf{NF}_t)$, where $\mathbf{NB}_t$ is the number of banks at t , $\mathbf{NF}_t$ is the number of firms at t , and A_{ijt} is the net exposure of i towards j at t . Recalling that creditors can be only banks, and debtors can be either firms or banks.

3. Methodology

3.1. Systemic risk computation

We compute the systemic risk on the exposure matrix $\mathbf{A}$ following the *differential DebtRank* (Bardoscia et al., 2015) methodology. The aggregate loan extended to j is A_j . At period 0, we impose an exogenous shock on agent j , reducing its equity by a fraction of ζ .⁶ It will cause a subsequent loss, at period 1, on their cred-

origin (earmarked or non-earmarked), interest rate, and risk classification of the operation and the client.

⁵ BM&FBOVESPA is a privately-owned company that was created in 2008 through the integration of the Sao Paulo Stock Exchange (Bolsa de Valores de Sao Paulo) and the Brazilian Mercantile & Futures Exchange (Bolsa de Mercadorias e Futuros). As Brazil's main intermediary for capital market transactions, the company develops, implements and provides systems for trading equities, equity derivatives, fixed-income securities, federal government bonds, financial derivatives, spot FX, and agricultural commodities.

⁶ In this paper, there are two-period notations. The subscript notation t – present in the previous section and omitted in this subsection for simplicity – refers to the

iters, indexed by i , whose aggregate value is equal to $A_j\zeta$. At period 2, j 's creditors will propagate this loss to their creditors similarly, and so on. We define $L_{ij}(t)$ as the accumulated loss transmitted by j to i up to period t . Moreover, $\Delta L_{ij}(t) = L_{ij}(t) - L_{ij}(t-1)$ is the new flow of loss transmitted by j to i and $L_i(t) = \sum_j L_{ij}(t)$ is the total loss transmitted to i by their debtors up to t . Finally, $\Delta L_i(t) = L_i(t) - L_i(t-1)$ is the variation in the total loss transmitted to i by their debtors up to t .

There are two mechanisms of the loss distribution. In the first mechanism, which corresponds to the standard approach of systemic risk computation in network models, the loss distribution is homogeneous, proportional to the loan extended by each creditor to the distressed debtor. In the second mechanism, the distribution is heterogeneous. Creditors are ranked according to a given heuristic. The distressed debtor defaults first on the top creditor of the rank, transmitting only the residual loss to the remaining creditors. The rest of this section discusses each mechanism more formally.

Homogeneous loss distribution. In the homogeneous case, the dynamics of loss propagation are represented by the following equations:

$$\Delta L_{ij}(t) = \min \left(A_{ij} - L_{ij}(t-1), A_{ij} \frac{[L_j(t-1) - L_j(t-2)]}{E_j} \right), \quad (1)$$

$$\Delta L_i(t) = \min \left(E_i - L_i(t-1), \sum_j \Delta L_{ij}(t) \right), \quad (2)$$

in which $t \geq 0$ and E_j is agent j 's equity. Thus, when an agent j suffers an additional loss equal to a fraction ζ of its equity, it will impose a loss to its creditors that corresponds to ζ times their exposures towards j . Observe that equity positions as well as the exposure network are time-invariant, i.e., they are taken as exogenous. The propagation considers stress differentials rather than stress absolute values (hence the methodology's name) to avoid double-counting.

Observe that, from Eq. 1, $L_{ij}(t)$ cannot be greater than A_{ij} , i.e., j cannot impose to i a loss greater than i 's exposures towards j . When $L_{ij}(t) = A_{ij}$, j stops imposing losses on i . Moreover, as can be observed from Eq. 2, $L_i(t) - L_i(t-1)$ – the loss imposed on agent i – cannot be greater than E_i . That is, i 's losses cannot be greater than its equity. When $L_i(t) = E_i$, i stops propagating losses to other agents.

Heterogeneous loss distribution. In the second approach, the distressed debtor prefers to default on certain creditors first. Agent j will rank their creditors as $i = 1, 2, \dots, J$, where J is j 's number of creditors. It prefers to default on creditor 1 first, then on creditor 2, and so on. Therefore, for any creditor i , j prefers to default on agent k rather than on i iff $k < i$. Debtor j transmits all the loss

date (month-year). The notation between parenthesis refers to the number of iterations after the shock, where 0 refers to the period in which the shock is imposed.

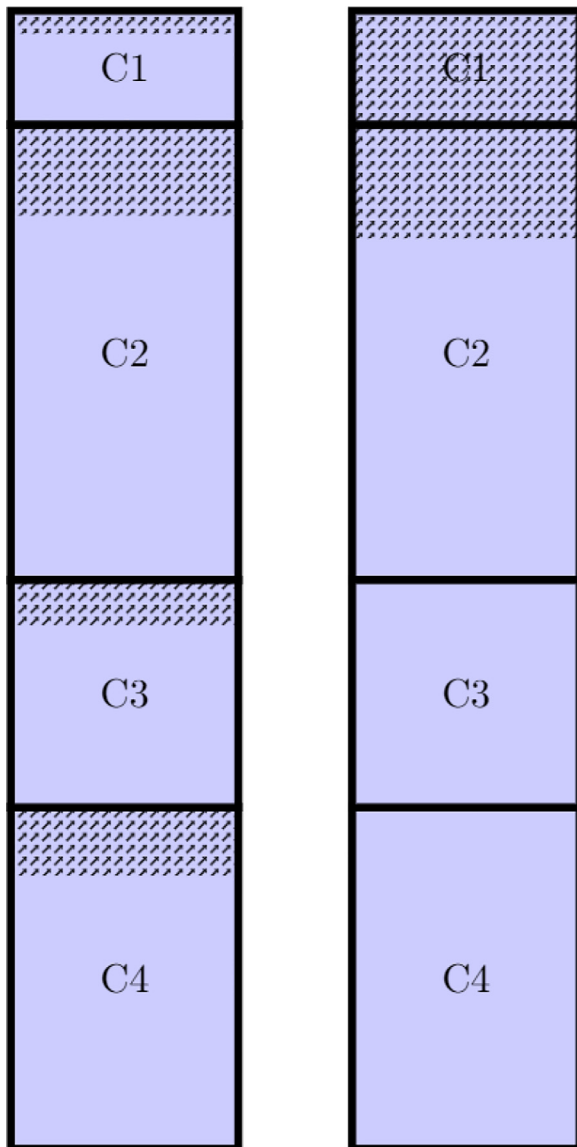


Fig. 1. Homogeneous (left) and heterogeneous (right) loss distribution. The loan extended by each creditor is proportional to the heights of the rectangles. The hatched area corresponds to the loss. In the homogeneous case, the fraction of the hatched area (0.2) is the same in all rectangles. The sum of the hatched area (i.e., the aggregate loss) is the same in both figures.

to creditor 1 at the limit of the loan extended by creditor 1 to j . Only the residual loss, if any, will be transmitted to creditor 2. The process continues until j has transmitted the entire loss to their creditors. Therefore, Eq. 1 is replaced by

$$\Delta L_{ij}(t) = \min \left(A_{ij} - L_{ij}(t-1), A_j \zeta - \sum_k \Delta L_{kj}(t) \right), \quad (3)$$

where $\zeta = \frac{L_j(t-1) - L_j(t-2)}{E_j}$ and $k = (1, \dots, i-1)$. Therefore, the loss propagated to a certain creditor i is equal to the aggregate loss agent j will transmit to its creditors, $A_j \zeta$, minus the loss already transmitted to agents $1, \dots, i-1$ – that is, the other agents j prefers to default on rather than on i . Eq. 2 also holds in this case.

Fig. 1 illustrates the differences between the two approaches. Suppose a distressed debtor has four creditors. They are ranked according to the debtor's default preference, meaning the debtor prefers to default on creditor 1 (C1) first. The loss the debtor will transmit to their creditors corresponds to 20% of its entire debt.

Table 2

Average systemic risk for different levels of ζ , weighted by the agent's equity. Legend: HOM: homogeneous loss distribution; RND: random sorting; EQT: equity heuristic; EAR: equity-to-assets ratio heuristic; KOU: out-degree heuristic; LOA: loan granted heuristic.

Type	Method	$\zeta=0.1$	$\zeta=0.5$	$\zeta=1.0$
Firms	HOM	0.20	0.87	1.43
	RND*	0.59	1.34	2.00
	EQT	0.89	1.42	2.01
	EAR	0.80	1.31	1.88
	KOU	1.03	1.55	2.16
	LOA	1.38	2.36	2.73
Banks	HOM	0.14	0.58	0.86
	RND*	0.42	0.80	1.11
	EQT	0.42	0.85	1.06
	EAR	0.36	0.64	0.96
	KOU	0.47	0.94	1.15
	LOA	0.73	1.13	1.30

*: Average of 10 realizations.

In the figure on the left, which corresponds to the homogeneous loss distribution case, the debtor transmits to each creditor a loss that corresponds to 20% of the loan granted by the creditor. In the heterogeneous case (figure on the right), the debtor transmits the maximum loss it can to C1 (in this case, all loans granted by C1) and only the residual loss to C2.

Systemic risk. After the initial shock on j , following one of the loss distribution mechanisms, the system converges after a sufficiently large number of periods $T \gg 1$. Then we have the final matrix of losses $\mathbf{L}^{j,\zeta} \in N \times 1$, where $L_i^{j,\zeta}$ is the total loss suffered by agent i after an initial shock of size ζ at agent j .

We repeat this process for the other agents. We define the systemic risk of agent i as

$$s_{i,\zeta} = 100 \times \frac{\sum_j [L_j^{i,\zeta} - L_j^{i,\zeta}(0)]}{\sum_j E_j}, \quad (4)$$

where $L_j^{i,\zeta}(0) = \zeta E_j$ if $j = i$ and 0 otherwise. Therefore, $s_{i,\zeta}$ measures the percentage of the aggregate agents' equity which is lost as a consequence of an initial shock of size ζ to agent i 's equity. Note that we remove the initial shock to the agent i ($L_i^{i,\zeta}(0)$) from the computation of the systemic risk, as we are interested only in the losses caused by the contagion. Moreover, we compute $s_{i,\zeta}$ also for the agent that suffered the initial shock. Due to network cyclicity, a shock propagated by a given agent can hit it back.

3.2. Machine learning techniques

After computing the systemic risk for our data instances using the methodology presented in Section 3.1, we will employ two machine learning techniques – random forest (RF) and XGBoost (XB) – to assess the determinants of the systemic risk (Section 4.2). The variable to be predicted is $s_{i,\zeta}$ (Eq. 4) and the explanatory variables are those presented in Table 3. Shapley values will be used to give a better interpretability to our results. These techniques will be discussed briefly in this subsection.

Random forest and XGBoost. RF (Breiman, 2001) and XB (Friedman, Hastie, Tibshirani et al., 2000) are ensemble learning methods that can be used for both classification and regression. RF operates by constructing several decision trees.⁷ For regression tasks, which is the case in this paper, it returns the average prediction of the individual decision trees. XB is an optimization algorithm that works with an ensemble of weak predictors (usually, decision trees) and creates a more efficient predictor model.

⁷ On decision trees, see, e.g., Breiman, Friedman, Stone, & Olshen (1984).

Table 3
Potential determinants assessed in the study.

Type	Variable	Acronym
Financial Variable	Equity (net worth)	NW
Financial Variable	Total borrowing-to-equity ratio	CA
Financial Variable	Dummy for the agent's type (firm: 1)	Type
Network Measurement (strictly local)	In-degree	Kin
Network Measurement (mixed)	PageRank	PR
Network Measurement (mixed)	Weighted clustering	WC
Network Measurement (mixed)	Weighted eigenvector centrality	EC
Network Measurement (mixed)	Strength	ST
Network Measurement (global)	Density of the financial network	dens

At each boosting stage, the XB algorithm attempts to increase the performance of the predecessor model by including a new estimator. In both cases, the purpose is to estimate a predicted output $\hat{y}_i$ from an observed output y_i and a vector of explanatory variables X_i .

The models are trained and validated through a process known as *repeated k-fold cross-validation*. The data set, comprised of the output to be predicted and a set of potential explanatory variables, is split into k different parts (folds). $k - 1$ folds are used in the development of the model. Then, the model is trained on the remaining fold: the predicted output $\hat{y}_i$ and the observed output y_i of the remaining fold are used to compute score measures, such as the root mean squared error (RMSE) and the R^2 . Each fold is used as the testing data set. Hence, for instance, in a repeated k -fold cross-validation with $k = 5$ and 10 repetitions, a total of 50 regressions are run.

These score measures are used to tune the number of estimators of both methods. In the RF, the number of estimators is the number of decision trees in each forest. In the XB, this is the number of boosting stages to be performed. The number of estimators varies within a grid of ascending values. For each of these values, the regressions are run, and the average score is computed. The number of estimators is chosen so that increasing it does not improve the performance of the method.

Shapley values. We go further on the interpretability of our results by resorting to the computation of Shapley values. This approach, originated from the coalition games theory (Shapley, 1953; Shoham & Leyton-Brown, 2008), provides evidence on features' importance. Moreover, Shapley values can also inform whether a given feature is positively or negatively correlated to the output. We compute Shapley values through the SHAP (SHapley Additive exPlanation) framework proposed by Lundberg & Lee (2017). The authors propose an explainer model g aiming at predicting an output using a set of M features as inputs. The predicted value for a given data instance is given by

$$g(z') = \phi_0 + \sum_{i=1}^M \phi_i z'_i, \quad (5)$$

where z' is a binary variable indicating whether feature i was included in the model or not. Therefore, the SHAP value ϕ_i indicates to what extent the feature i shifts the predicted value up or down from a given mean output ϕ_0 . Lundberg & Lee (2017) showed, under certain properties (local accuracy, missingness, and consistency), ϕ_i corresponds to the Shapley value of the game theory. The SHAP value of feature i is given by

$$\phi_i = \sum_{S \subseteq M \setminus \{i\}} \frac{|S|!(|M| - |S| - 1)!}{M!} [F(S \cup \{i\}) - F(S)]. \quad (6)$$

Therefore, the SHAP value of feature i for a given data instance computes the difference between the predicted value of the instance using all features in S plus feature i , $F(S \cup \{i\})$, and the prediction excluding feature i , $F(S)$. This is weighted and summed

over all possible feature vector combinations of all possible subsets S .⁸

4. Results

4.1. General results

We compute the systemic risk (Eq. 4) considering three values of ζ (0.1, 0.5, and 1.0) for 7076 observations (2,060 date-bank data instances and 5016 date-firm data instances). We consider five loss distribution mechanisms:

- The homogeneous loss distribution mechanism.
- The heterogeneous distribution mechanism according to the four heuristics discussed in Section 1: equity, equity-to-assets ratio, out-degree, and loan extended. For instance, according to the equity heuristic, the distressed debtor ranks their creditors in ascending order by their equities, preferring to default on those with smaller equity first. The other heuristics follow the same reasoning (i.e., creditors are ranked in ascending order according to their equity-to-assets ratio, out-degree, or their loan extended).
- The heterogeneous distribution mechanism in which the creditors are randomly sorted. This is to check whether the differences regarding the homogeneous case (if any) are due to the heterogeneous distribution *per se* or also due to the heuristic adopted. We run 10 different realizations considering this loss distribution mechanism.

The results are presented in Figs. 2 and 3. It can be seen the systemic risk increases when a heterogeneous loss distribution is adopted. Moreover, the systemic risk of firms is smaller than that of banks. This is because firms are only debtors in the network. They cannot be hit back by shocks to themselves. By contrast, banks can be both debtors and creditors. Thus, shocks to banks are amplified by the cyclicity of the network.

The heuristic also plays an important role in the increase of the systemic risk. When loss is transmitted preferentially to more fragile agents rather than randomly, the average systemic risk is higher. The loan granted heuristic entails the highest average systemic risk. Moreover, the rise in the systemic risk brought by the heterogeneous distribution over the homogeneous case decreases with the level of the initial shock. For an initial shock of 0.1, the ratio between the systemic risk entailed by the heterogeneous distribution of losses (random sorting, equity, equity-to-assets ratio, out-degree, and loans granted) and that entailed by the standard approach is, respectively, 7.4, 11.4, 15.4, 14.0, and 27.5 for the banks. If the initial shock is 0.5, these ratios decrease to 3.0, 3.7, 3.9, 4.6, and 7.3. For an initial shock of 1.0 (complete default), these values are even smaller: 2.4, 2.5, 2.8, 3.2, and 4.6. A similar decreasing pattern can be observed for firms. Finally, the heuristics also

⁸ For details on the calculation of SHAP values, see, e.g., Lundberg & Lee (2017) and Kalair & Connaughton (2021).

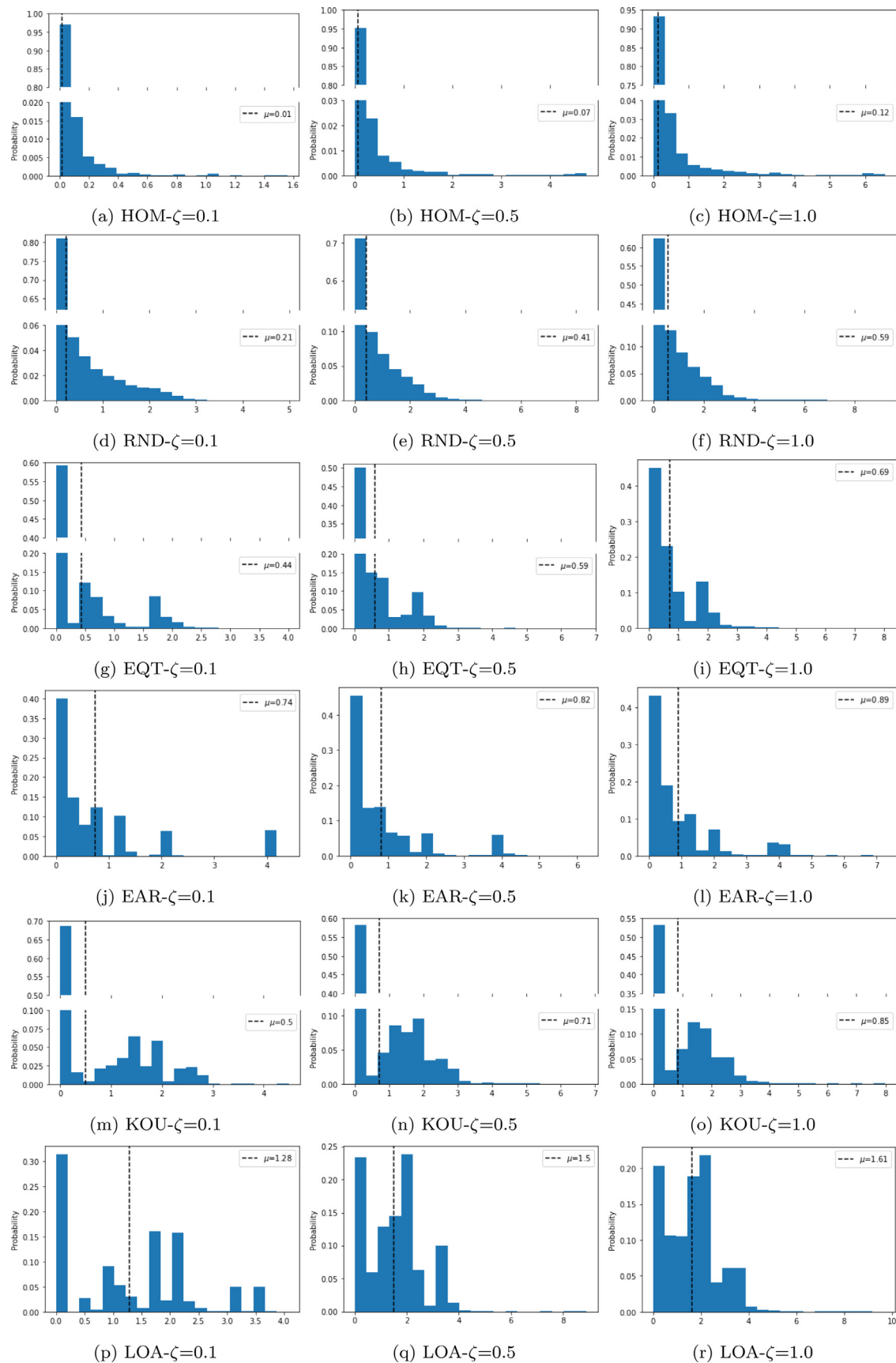


Fig. 2. Density probability of the systemic risk – firms. Legend: HOM: homogeneous loss distribution; RND: random sorting; EQT: equity heuristic; EAR: equity-to-assets ratio heuristic; KOU: out-degree heuristic; LOA: loan granted heuristic. The vertical lines indicate the average systemic risk.

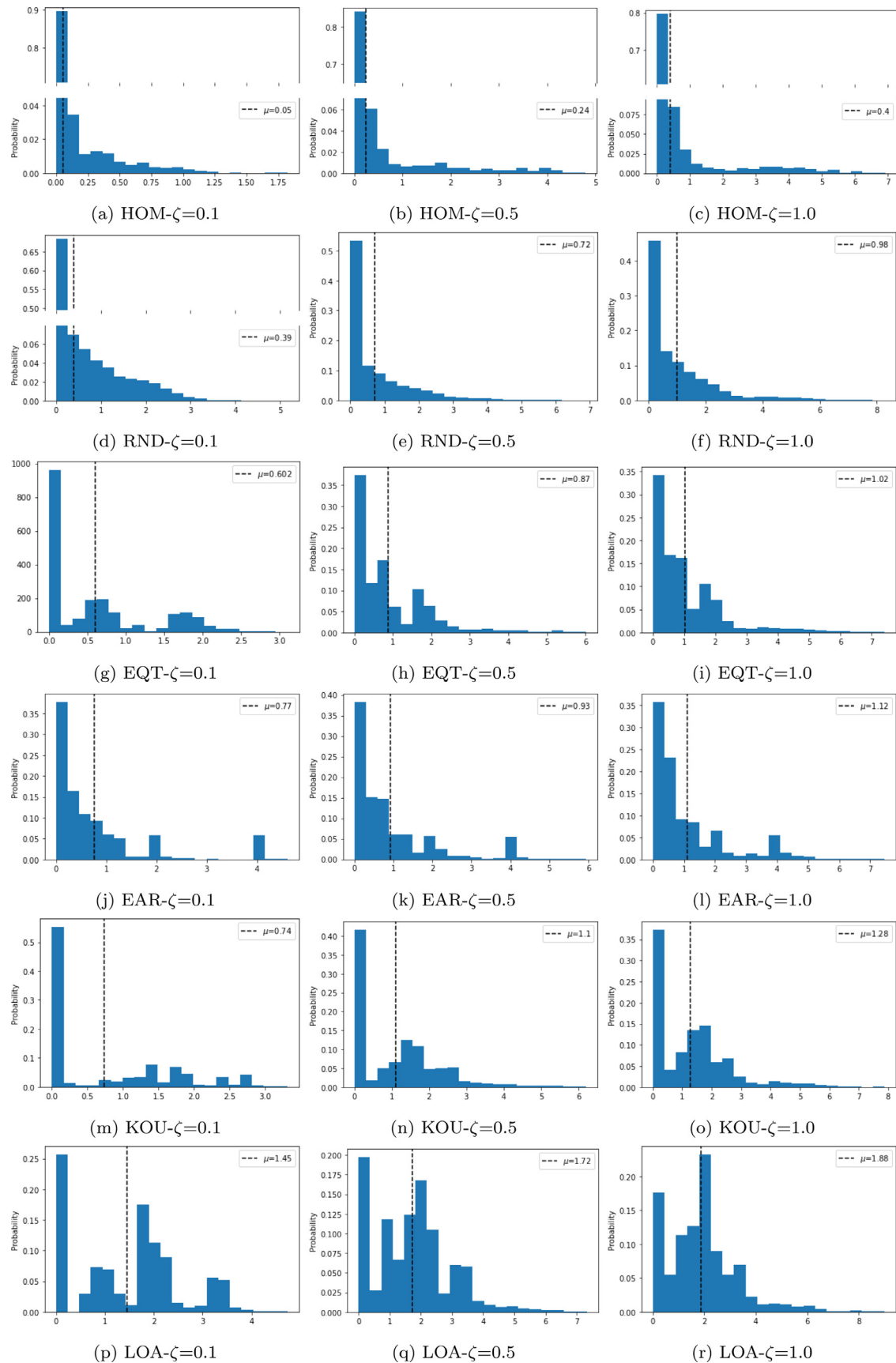


Fig. 3. Density probability of the systemic risk – banks. Legend: HOM: homogeneous loss distribution; RND: random sorting; EQT: equity heuristic; EAR: equity-to-assets ratio heuristic; KOU: out-degree heuristic; LOA: loan granted heuristic. The vertical lines indicate the average systemic risk.

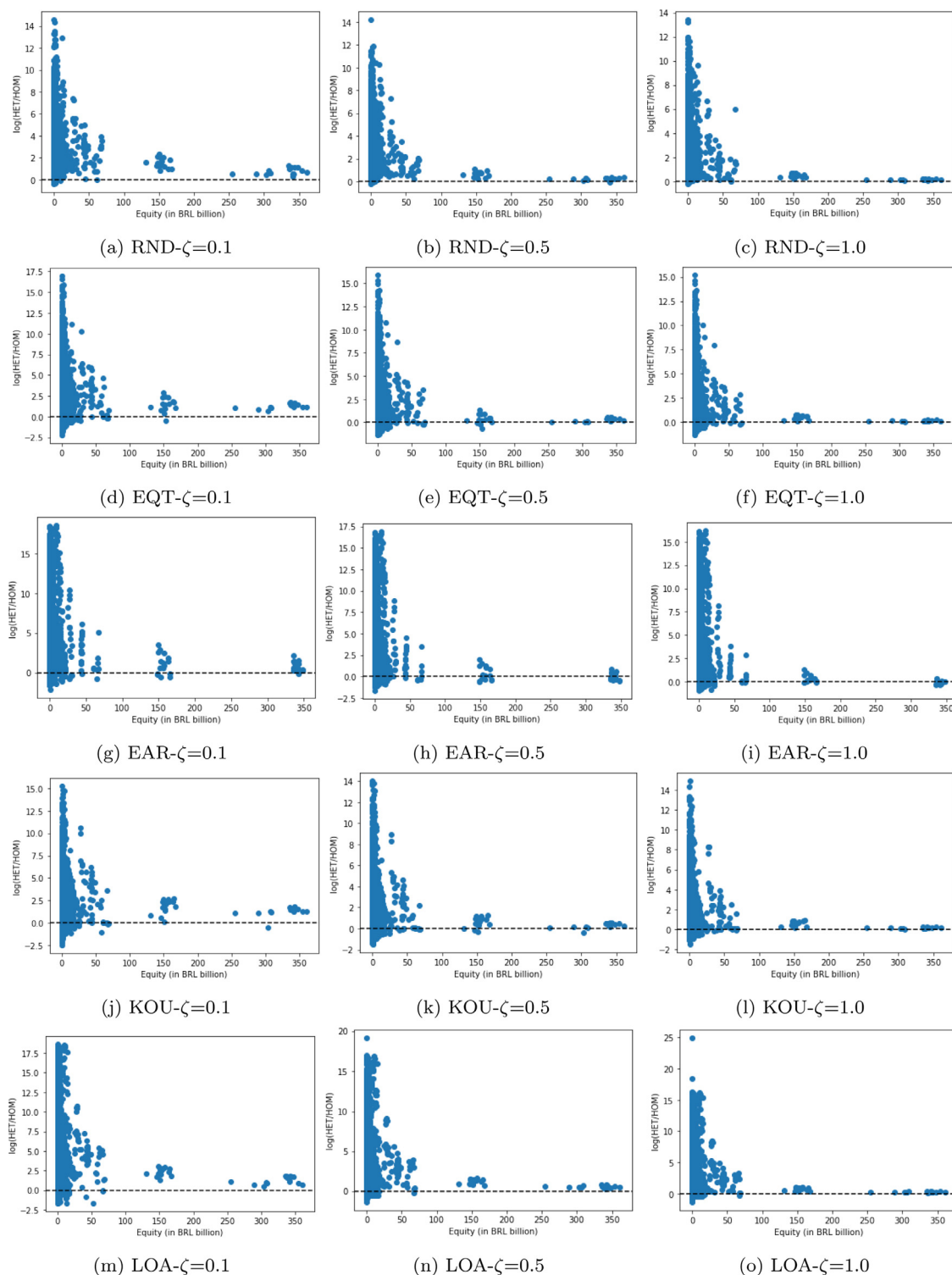


Fig. 4. Systemic risk entailed by the heterogeneous loss distribution-to-systemic risk entailed by the homogeneous loss distribution ratio – firms. Legend: RND: random sorting; EQT: equity heuristic; EAR: equity-to-assets ratio heuristic; KOU: out-degree heuristic; LOA: loan granted heuristic.

change the distribution of the systemic risk. The distribution of the systemic risk entailed by the homogeneous distribution and the random sorting heterogeneous distribution of losses is right-skewed. When some heuristic is introduced, the distribution of the systemic risk becomes multimodal.

Table 2 presents the weighted average systemic risk $\sum_i w_{it} s_{i\zeta t}$, where w_{it} is the agent i 's participation in total equity at period t .

Thus, this is the loss caused by a shock in the agents weighted by their equities. In this case the differences between the systemic risk entailed by the different approaches are less remarkable. This suggests the increase in the systemic risk brought by the heterogeneous loss distribution depends on the agents' equity. We then investigate the relationship between the systemic risk entailed by the heterogeneous distributions of losses and that entailed by the

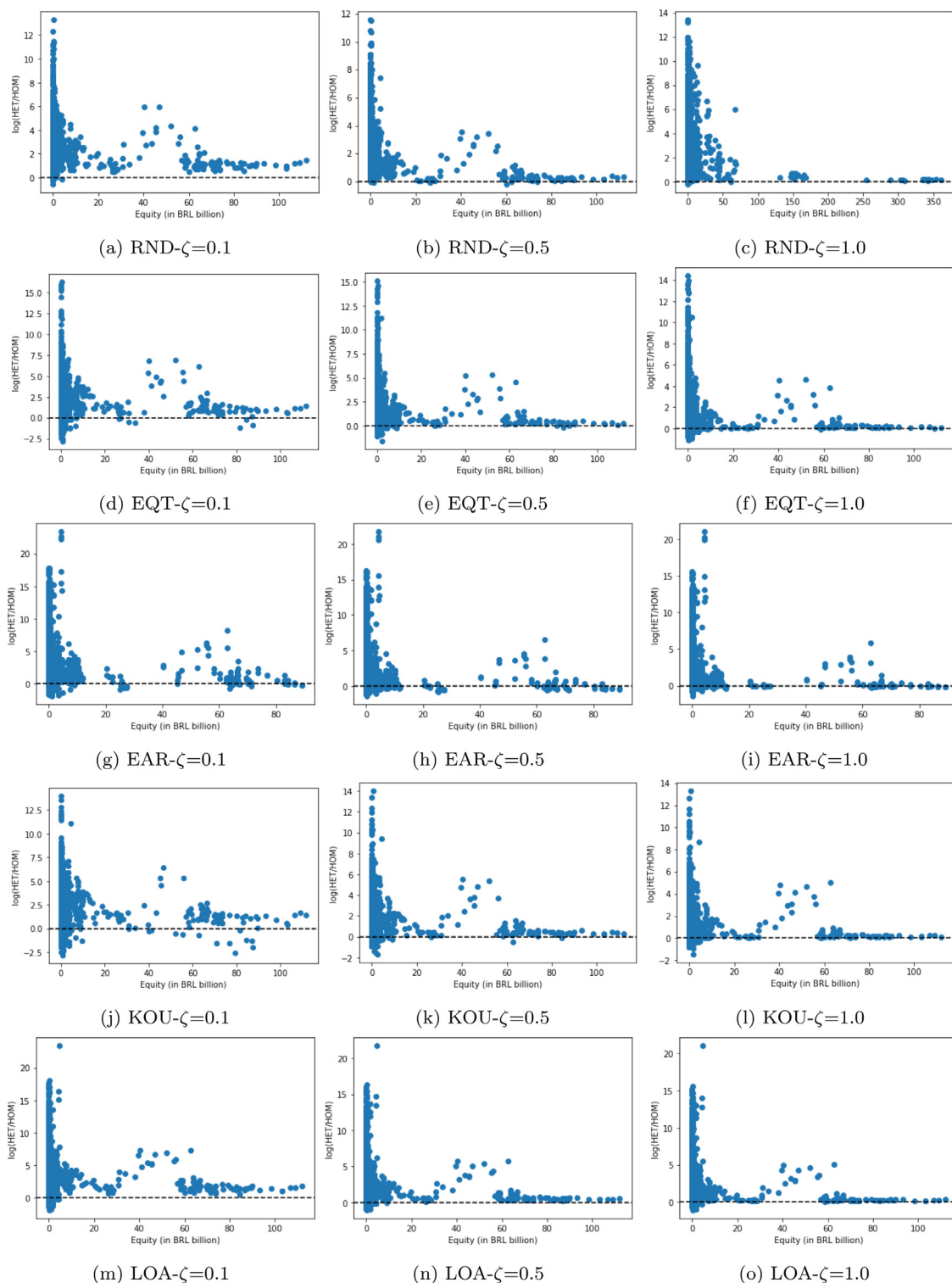


Fig. 5. Systemic risk entailed by the heterogeneous loss distribution-to-systemic risk entailed by the homogeneous loss distribution ratio – banks. Legend: RND: random sorting; EQT: equity heuristic; EAR: equity-to-assets ratio heuristic; KOU: out-degree heuristic; LOA: loan granted heuristic.

homogeneous distribution HET/HOM at the agent level. Results are presented in Figs. 4 and 5. For the sake of better visualization this ratio is presented on a natural logarithmic scale. The highest differences between the systemic risk entailed by the heterogeneous loss distributions and that entailed by the homogeneous approach are observed in the small-sized agents. In a few cases, this ratio is smaller than one.

4.2. Risk-sharing versus shock propagation

Our results can be interpreted in the light of the dual role of financial networks. They can work as a channel for risk-sharing, but also for shock propagation (Chinazzi & Fagiolo, 2015). The role played by interconnectedness in the systemic risk depends on which effect prevails over the other. A more interconnected net-

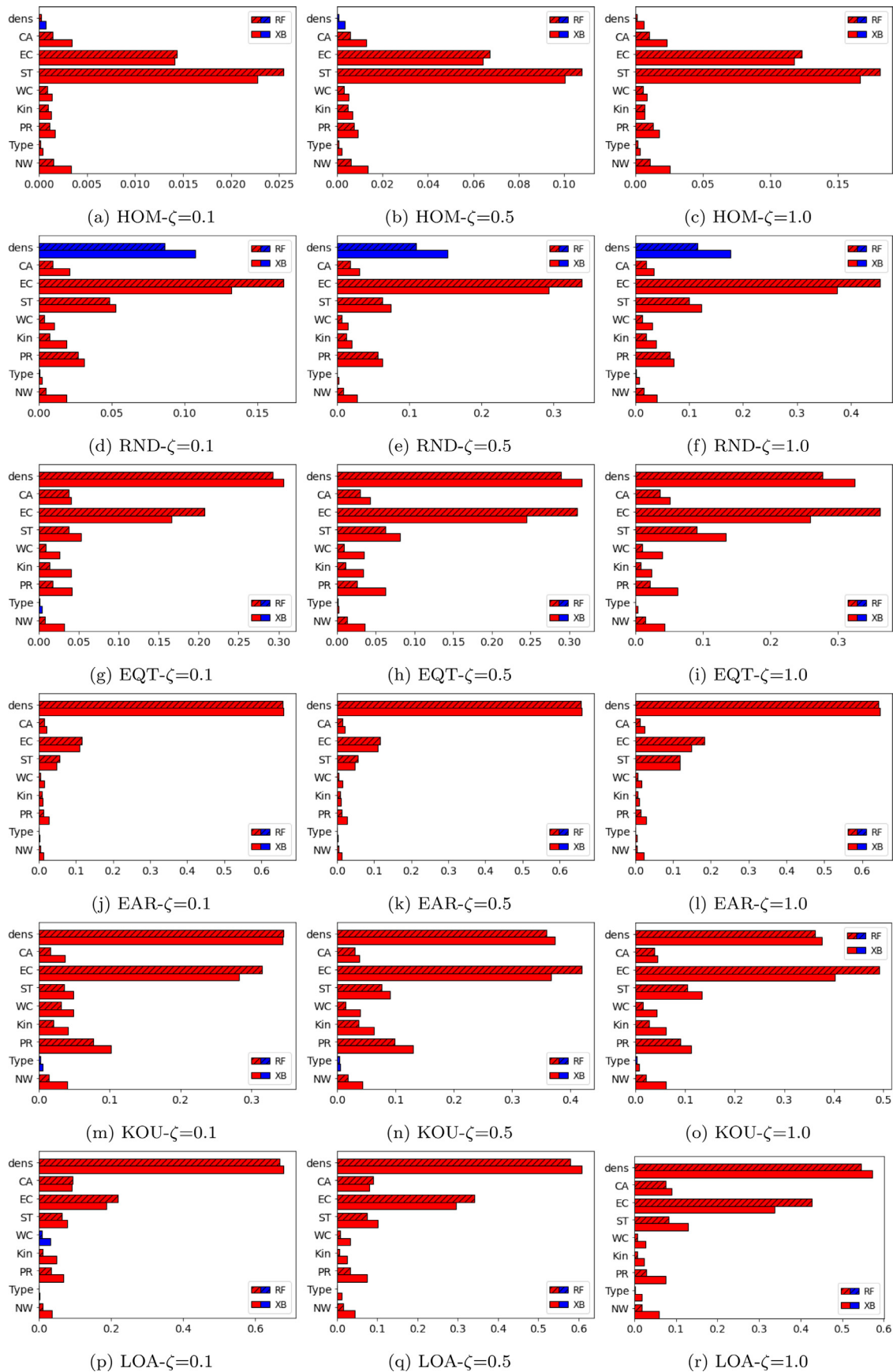


Fig. 6. Average absolute SHAP values. Legend: HOM: homogeneous loss distribution; RND: random sorting; EQT: equity heuristic; EAR: equity-to-assets ratio heuristic; KOU: out-degree heuristic; LOA: loan granted heuristic. Bars in red (blue) indicates the corresponding feature is positively (negatively) correlated to the SHAP value. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

work will be more robust if the risk-sharing effect prevails over the shock propagation effect. Hence, the systemic risk is smaller. Otherwise, more interconnections will increase the level of systemic risk. An interesting question concerning this issue is the following: does the DPO affect the balance between both risk-sharing and shock propagation channels in a financial network?

We go further on this hypothesis by assessing the effect of interconnectedness on our measures of systemic risk. We employed two machine learning techniques – RF and XB – to predict the agents' systemic risk. We use two sets of predictive attributes: financial variables (balance sheet ratios and other financial information) and topological measures borrowed from the complex networks literature. The potential explanatory variables are described in Table 3. For the first set of variables, we use the economic agent's equity, total borrowing-to-equity ratio, and a dummy variable for firms. For the second set of variables, we follow Silva & Zhao (2016) and select at least one strictly local, mixed, and global network measurement to capture local to global characteristics of the network structure comprehensively. We select the unweighted PageRank (mixed measure), which is the most influential driver of Brazilian banks' systemic risk (Alexandre, Silva, Connaughton, & Rodrigues, 2021). We add other three weighted mixed measures: the weighted clustering coefficient, pointed by the literature as an important indicator of systemic risk (Castellano, Cerqueti, Clemente, & Grassi, 2021; Cerqueti, Clemente, & Grassi, 2021; Minoiu & Reyes, 2013; Tabak, Takami, Rocha, Cajueiro, & Souza, 2014), the weighted eigenvector centrality, and the strength. We also introduce the in-degree (strictly local) to capture direct loss transmission. Finally, we add the network density (global measure) as a proxy for interconnectedness. For a given date, we calculate the density of the financial network as follows:

$$dens_t = \frac{k_t}{NB_t \times (NB_t + NF_t - 1)}, \quad (7)$$

where k_t is the total number of links of the network at period t , NB_t is the total number of different banks in the network at t , and NF_t is the total number of different firms. The denominator represents the maximum number of links, as only banks can be lenders (except for themselves), but both banks and firms can be borrowers. Thus, the density is the number of links over the number of possible links.

We implemented a repeated k -folds cross-validation with $k = 5$ folds and 10 repetitions. After tuning the number of estimators of both methods using the root mean squared error (RMSE) as the score measure within the grid [30, 50, 70, 100, 300, 500], we set the value of both parameters as 50. Finally, we apply both techniques to predict the systemic risk. The outputs to be predicted are the six measures of systemic risk, each one engendered by a different methodology of loss distribution (homogeneous, random sorting, equity heuristic, equity-to-assets heuristic, out-degree heuristic, and loan granted heuristic). The potential explanatory variables are those presented in Table 3.

We assess the importance of each feature in driving the systemic risk by computing Shapley values through the SHAP (SHapley Additive exPlanation) framework (Lundberg & Lee, 2017). Once the two machine learning techniques are employed as explainer models for our systemic risk measures, we compute the SHAP values. Then we compute, for each feature presented in Table 3, the average absolute SHAP value over all data instances. Finally, we multiply this value by the sign of the correlation between the feature value and the SHAP value. The final value gives us two pieces of information. Its absolute value shows the feature importance in driving the output. Its sign informs whether the feature is positively or negatively correlated to the output.

The results are presented in Fig. 6. Density has a meagre effect in driving systemic risk under the homogeneous loss distribu-

tion rule. However, this is an important systemic risk driver when the heterogeneous distribution of loss is adopted, rivaling in importance with the weighted eigenvector centrality. When creditors are sorted randomly, the impact of interconnectedness is negative and smaller in absolute value than that of the weighted eigenvector centrality. On the other hand, when any of the four heuristics (equity, equity-to-assets ratio, out-degree, and loan granted) is adopted, its impact is positive. This means that a more interconnected network leads to a higher systemic risk when losses are mainly transmitted to fragile creditors.

These results corroborate the hypothesis that the DPO affects the balance between the risk-sharing and shock propagation channels. Under the homogeneous distribution of loss, interconnectedness (as measured by the network's density) has no significant impact on systemic risk. This suggests both effects (risk-sharing and shock propagation) almost counterbalance. However, when losses are transmitted mostly to fragile creditors, the shock propagation effect gains relative importance over the risk-sharing effect. In this case, higher interconnectedness leads to higher systemic risk, as the financial network acts mainly as a channel for shock propagation rather than risk sharing.

5. Concluding remarks

In this paper, we computed the systemic risk relaxing the assumption of pro-rata distribution of losses by distressed debtors among their creditors. This assumption is adopted by traditional network models of systemic risk. However, this assumption is not realistic. Empirical studies show debtors are more likely to default on fragile creditors. Default implies the impairment of the debtor-creditor relationship. For this reason, debtors prefer to default first on those creditors whose relationships are less valuable to them in the future.

For a Brazilian multilayer credit network (interbank credit network and bank-firm credit network), we compared the systemic risk computed under the homogeneous and heterogeneous rule of the loss distribution. In the homogeneous rule, distressed debtors transmit the loss to their creditors proportionally to the loan extended by these creditors. In the heterogeneous rule, creditors are ranked in ascending order according to some criterion (equity, equity-to-assets ratio, out-degree, loan extended, or randomly). Once distressed debtors compute the aggregate loss to be transmitted to their creditors, they default on their creditors following this default pecking order. Therefore, under the heterogeneous rule (except in the random sorting case), losses are mostly transmitted to the more fragile creditors.

Our results show systemic risk increases substantially when the heterogeneous loss distribution is adopted. The rise in the systemic risk brought by the heterogeneous distribution over the homogeneous case decreases with the level of the initial shock and is higher for small-sized agents. These results are related to the dual role of financial networks. They can be a channel for both risk-sharing and shock transmission. It is possible that the DPO affects the balance between the two effects, in such a way the shock transmission effect of the financial network gains relative importance over the risk-sharing effect, and the systemic risk increases.

We tested this hypothesis by assessing the determinants of systemic risk under different rules of the loss distribution. Among the potential explanatory variables, we included the density of the financial network. We performed this task employing two machine learning techniques (random forest and XGBoost) and Shapely values to give more interpretability to our results. The results corroborated this hypothesis. Under the homogeneous rule, the two effects of the financial network (risk-sharing and shock transmission) counterbalance and the density has a meagre impact on systemic risk. On the other hand, when losses are mostly transmit-

ted to more fragile creditors, the density has a positive impact on systemic risk, suggesting in this case the financial network acts mainly as a shock transmission channel.

Our study contributes to the literature on systemic risk by showing the magnitude of the systemic risk depends on the strategy of loss distribution adopted by the debtors. The unrealistic assumption that losses are evenly distributed among creditors leads to an underestimation of systemic risk. Moreover, we shed new light on the role played by interconnectedness in systemic risk. When the assumption of homogeneous loss distribution is relaxed, the shock transmission channel of the financial network becomes relatively more important *vis-à-vis* the risk-sharing channel. In this case, interconnectedness has a positive impact on systemic risk, as the financial network is working mainly as a shock transmission channel.

Acknowledgements

T.C.S. acknowledges CNPq, Brazil (Grant No. 302703/2022-5) for the financial support given for his research. F. A. R. acknowledges CNPq, Brazil (Grant No. 309266/2019-0) for the financial support given for his research.

References

- Acemoglu, D., Ozdaglar, A., & Tahbaz-Salehi, A. (2015). Systemic risk and stability in financial networks. *American Economic Review*, 105(2), 564–608.
- Alexandre, M., Silva, T. C., Connaughton, C., & Rodrigues, F. A. (2021). The drivers of systemic risk in financial networks: Adata-driven machine learning analysis. *Chaos, Solitons & Fractals*, 153, 111588.
- Asarnow, E., & Edwards, D. (1995). Measuring loss on defaulted bank loans: A 24-year study. *Journal of Commercial Lending*, 77(7), 11–23.
- Asimakopoulou, I., Avramidis, P. K., Malliaropoulos, D., & Travlos, N. (2016). Moral hazard and strategic default: Evidence from greek corporate loans. *Bank of Greece Working Paper*, 211.
- Bardoscia, M., Battiston, S., Caccioli, F., & Caldarelli, G. (2015). Debtrank: A microscopical foundation for shock propagation. *PloS One*, 10(6), e0130406.
- Battiston, S., Gatti, D. D., Gallegati, M., Greenwald, B., & Stiglitz, J. E. (2012a). Liaisons dangereuses: Increasing connectivity, risk sharing, and systemic risk. *Journal of Economic Dynamics and Control*, 36(8), 1121–1141.
- Battiston, S., Puliga, M., Kaushik, R., Tascia, P., & Caldarelli, G. (2012b). Debtrank: Too central to fail? financial networks, the FED and systemic risk. *Scientific Reports*, 2, 541.
- Bertschinger, N., Hoefler, M., & Schmand, D. (2019). Strategic payments in financial networks. *arXiv preprint arXiv:1908.01714*.
- Bonaccolto, G., Caporin, M., & Maillet, B. B. (2022). Dynamic large financial networks via conditional expected shortfalls. *European Journal of Operational Research*, 298(1), 322–336.
- Bond, P., & Rai, A. S. (2009). Borrower runs. *Journal of Development Economics*, 88(2), 185–191.
- Bottazzi, G., De Sanctis, A., & Vanni, F. (2020). Non-performing loans and systemic risk in financial networks. Available at SSRN, 3539741.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- Breiman, L., Friedman, J., Stone, C. J., & Olshen, R. A. (1984). *Classification and regression trees*. CRC press.
- Breza, E. (2012). Peer effects and loan repayment: Evidence from the krishna default crisis. *Job Market Paper MIT*, 1.
- Caccioli, F., Catanach, T. A., & Farmer, J. D. (2012). Heterogeneity, correlations and financial contagion. *Advances in Complex Systems*, 15(supp02), 1250058.
- Carrasco, V., & Salgado, P. (2014). Coordinated strategic defaults and financial fragility in a costly state verification model. *Journal of Financial Intermediation*, 23(1), 129–139.
- Castellano, R., Cerqueti, R., Clemente, G. P., & Grassi, R. (2021). An optimization model for minimizing systemic risk. *Mathematics and Financial Economics*, 15(1), 103–129.
- Cerqueti, R., Clemente, G. P., & Grassi, R. (2021). Systemic risk assessment through high order clustering coefficient. *Annals of Operations Research*, 299(1), 1165–1187.
- Chinazzi, M., & Fagiolo, G. (2015). Systemic risk, contagion, and financial networks: A survey. *Institute of Economics, Scuola Superiore Sant'Anna, Laboratory of Economics and Management (LEM) Working Paper Series*, (2013/08).
- Das, S. R., & Meadows, R. (2013). Strategic loan modification: An options-based response to strategic default. *Journal of Banking & Finance*, 37(2), 636–647.
- Degryse, H., Nguyen, G., et al. (2007). Interbank exposures: An empirical examination of contagion risk in the belgian banking system. *International Journal of Central Banking*, 3(2), 123–171.
- Drozdz, L., & Serrano-Padial, R. (2018). Financial contracting with enforcement externalities. *Journal of Economic Theory*, 178, 153–189.
- D'Erasmo, P., & Mendoza, E. G. (2021). History remembered: Optimal sovereign default on domestic and external debt. *Journal of Monetary Economics*, 117, 969–989.
- Eisenberg, L., & Noe, T. H. (2001). Systemic risk in financial systems. *Management Science*, 47(2), 236–249.
- Elsinger, H., Lehar, A., & Summer, M. (2006). Risk assessment for banking systems. *Management Science*, 52(9), 1301–1314.
- Friedman, J., Hastie, T., Tibshirani, R., et al. (2000). Additive logistic regression: A statistical view of boosting (with discussion and a rejoinder by the authors). *Annals of Statistics*, 28(2), 337–407.
- Gai, P., & Kapadia, S. (2010). Contagion in financial networks. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 466(2120), 2401–2423.
- Guiso, L., Sapienza, P., & Zingales, L. (2013). The determinants of attitudes toward strategic default on mortgages. *The Journal of Finance*, 68(4), 1473–1515.
- Gupta, A., Wang, R., & Lu, Y. (2021). Addressing systemic risk using contingent convertible debt—a network analysis. *European Journal of Operational Research*, 290(1), 263–277.
- Halaj, G., & Kok, C. (2013). Assessing interbank contagion using simulated networks. *Computational Management Science*, 10(2–3), 157–186.
- Hu, J., Yang, L., & Yung-Ho, C. (2004). Ownership and non-performing loans: Evidence from taiwan's banks. *Developing Economics*, 42(3), 405–420.
- Kalair, K., & Connaughton, C. (2021). Dynamic and interpretable hazard-based models of traffic incident durations. *arXiv preprint arXiv:2102.08729*.
- Karthik, L., Subramanyam, M., Shrivastava, A., & Joshi, A. (2018). Prediction of wilful defaults: An empirical study from indian corporate loans. *International Journal of Intelligent Technologies & Applied Statistics*, 11(1).
- Li, S., Liu, Y., & Deininger, K. (2013). How important are endogenous peer effects in group lending? estimating a static game of incomplete information. *Journal of Applied Econometrics*, 28(5), 864–882.
- Li, S., Yanyan, L., & Deininger, K. (2009). How important are peer effects in group lending? *International Food Policy Research Institute (IFPRI). Discussion Paper*, 940.
- Lundberg, S., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *arXiv preprint arXiv:1705.07874*.
- Luu, D. T., & Lux, T. (2019). Multilayer overlaps and correlations in the bank-firm credit network of spain. *Quantitative Finance*, 19(12), 1953–1974.
- Minoiu, C., & Reyes, J. A. (2013). A network analysis of global banking: 1978–2010. *Journal of Financial Stability*, 9(2), 168–184.
- Nier, E., Yang, J., Yorulmazer, T., & Alentorn, A. (2007). Network models and financial stability. *Journal of Economic Dynamics and Control*, 31(6), 2033–2060.
- Rajan, R., & Dhal, S. (2003). Non-performing loans and terms of credit of public sector banks in india: An empirical assessment. *Reserve Bank of India Occasional Papers*, 24(3), 81–121.
- Roukny, T., Bersini, H., Piroette, H., Caldarelli, G., & Battiston, S. (2013). Default cascades in complex networks: Topology and systemic risk. *Scientific Reports*, 3, 2759.
- Salas, V., & Saurina, J. (2002). Credit risk in two institutional regimes: Spanish commercial and savings banks. *Journal of Financial Services Research*, 22(3), 203–224.
- Schiantarelli, F., Stacchini, M., & Strahan, P. E. (2020). Bank quality, judicial efficiency, and loan repayment delays in italy. *The Journal of Finance*, 75(4), 2139–2178.
- Shapley, L. S. (1953). A value for n-person games. *Contributions to the Theory of Games*, 2(28), 307–317.
- Shoham, Y., & Leyton-Brown, K. (2008). *Multiagent systems: Algorithmic, game-theoretic, and logical foundations*. Cambridge University Press.
- Silva, T. C., & Zhao, L. (2016). *Machine learning in complex networks*. Springer Publishing Company.
- de Souza, S. R. S., Silva, T. C., Tabak, B. M., & Guerra, S. M. (2016). Evaluating systemic risk using bank default probabilities in financial networks. *Journal of Economic Dynamics and Control*, 66, 54–75.
- Stern, G. H., & Feldman, R. J. (2004). *Too big to fail: The hazards of bank bailouts*. Brookings Institution Press.
- Summe, K. A. (2012). An examination of lehman brothers' derivatives portfolio post-bankruptcy – would dodd-frank have made a difference? In *Bankruptcy not bailout* (pp. 85–129). Hoover Institution, Stanford University.
- Tabak, B. M., Takami, M., Rocha, J. M., Cajueiro, D. O., & Souza, S. R. (2014). Directed clustering coefficient as a measure of systemic risk in complex banking networks. *Physica A: Statistical Mechanics and its Applications*, 394, 211–216.
- Tang, Q., Tong, Z., & Xun, L. (2022). Insurance risk analysis of financial networks vulnerable to a shock. *European Journal of Operational Research*, 301(2), 756–771.
- Tran, N.-K., Vuong, T., & Zeckhauser, R. J. (2018). Loss sequencing in banking networks: Threatened banks as strategic dominoes.
- Trautmann, S. T., & Vlahu, R. (2013). Strategic loan defaults and coordination: An experimental analysis. *Journal of Banking & Finance*, 37(3), 747–760.
- Upper, C. (2011). Simulation methods to assess the danger of contagion in interbank markets. *Journal of Financial Stability*, 7(3), 111–125.
- Vlahu, R. (2008). Collective strategic defaults: Bailouts and repayment incentives. In *Efa 2008 athens meetings paper*.